# Advanced Database

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## 1.

A resource and cost management plan is important in any IT project because it helps control how people, tools, and money are used during development. In an e-commerce application, the database system is responsible for storing product information, customer accounts, orders, and payment records. Because this data is very important, the project must be carefully planned.

First, the project manager identifies the resources needed for the database project. These resources may include database developers, database administrators, system analysts, and testers. Each team member is given specific tasks depending on their skills. For example, database developers design the database tables and relationships, while testers check if the database works correctly with the application.

The cost management plan includes different expenses such as employee salaries, database software licenses, cloud storage, and server hosting. The project manager prepares a budget at the beginning of the project and records all expenses during development.

Regular monitoring helps the team track spending and avoid going over the budget. This type of planning ensures the project runs smoothly and the database system is completed successfully.

## 2.

Resource leveling is a project management technique used to balance how resources are used in a project. In many projects, some workers may be given too many tasks while others have fewer responsibilities. Resource leveling helps solve this problem by adjusting the project schedule.

When resource leveling is used, tasks may be delayed or moved so that work is shared more evenly between team members. This ensures that no one is overloaded with too much work at the same time.

For example, if one database developer is assigned several tasks such as database design, query writing, and testing at the same time, the project manager may move some tasks to another time or give them to another team member.

Resource leveling improves project performance because it helps team members work more efficiently. It also reduces stress and helps improve the quality of the final project.

### 3.

A cost tracking sheet in Excel is a simple tool used to monitor the expenses of a project. It helps project managers compare the planned budget with the actual money spent during the project.

In Excel, a cost tracking sheet usually includes columns such as task name, estimated cost, actual cost, and cost difference. At the beginning of the project, the estimated cost for each activity is entered into the sheet. As the project continues, the actual costs are added.

Excel formulas can automatically calculate the difference between the planned cost and the real cost. This helps the project manager quickly see if the project is spending more money than expected.

Using Excel for cost tracking is very helpful because it organizes financial information clearly. It also helps project managers make better decisions and ensures that the project stays within the approved budget.

### 4.

A resource allocation plan explains how project resources will be assigned to different tasks during the project. The main purpose of this plan is to ensure that the right people and tools are available when they are needed.

In a database development project, the plan may assign database designers to create the database structure, developers to write queries and stored procedures, and testers to check database performance and security.

This plan also helps avoid resource conflicts. For example, two tasks should not require the same person at the same time. The project manager schedules tasks carefully so that work is shared fairly among team members.

A good resource allocation plan helps improve productivity and ensures that tasks are completed on time. It also allows the project manager to monitor progress and adjust resources if any problems occur during the project.